

SHIFT:LLM-powered If-Then feedback tracker for short-form video self-regulation

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Abstract

Short-form videos offer advantages of rapid information acquisition and intuitive content consumption. However, the combination of brevity and infinite scroll structures exacerbates habitual overuse problems. Existing intervention studies have failed to achieve fundamental behavioral change by relying on simple usage limits and one-time notifications, taking a uniform approach that does not consider individual viewing contexts. To overcome these limitations, this study proposes SHIFT, which applies If-Then theory. SHIFT guides users to establish specific plans of “if situation X occurs, then I will do action Y” and tracks actual execution to induce automatic behavioral change. The system collects real-time scroll patterns, automatically classifies viewing content using Vision-Language Models, and generates personalized intervention messages through LLM-based four distinct prompting strategies. A 4-week user study confirmed a 50% reduction in average daily usage time ($p < 0.001$) and achieved an 83% ($p < 0.001$) intervention success rate.

CCS Concepts

• **Human-centered computing** → **User studies; Interaction design theory, concepts and paradigms; Empirical studies in HCI**; • **Applied computing** → **Health informatics**.

Keywords

If-Then Planning, Short-form Video, Self-regulation, Personal Informatics, Large Language Model, Human-Computer Interaction

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1 Introduction

Amid rapid changes in the mobile media ecosystem, short-form platforms such as TikTok, Instagram Reels, and YouTube Shorts are fundamentally transforming everyday media consumption patterns through content lasting just seconds [4]. These platforms are designed for entertainment and maximize engagement time, enabling users to rapidly and consecutively view diverse topics [9]. However, users become absorbed in videos due to the infinite scroll UI of short-form video platforms and spend much longer time on short-form viewing than intended [18]. The key issue here is that most users do not accurately perceive their actual viewing time and content volume [30]. While users cognitively know they should stop scrolling, they continue in most cases due to the design characteristics of infinite scroll [7]. Consequently, existing digital intervention research and applications have primarily relied on behaviorist approaches such as scroll restrictions [2, 3, 8, 16].

Therefore, this study applied If-Then planning [10]. This theory can improve goal achievement rates by establishing specific action plans in advance. Additionally, LLMs (Large Language Models) can be utilized to analyze viewing patterns (short-form video viewing records, total viewing time, viewing time periods) and propose alternative plans, thereby inducing autonomous behavioral change [19]. Based on this, we propose an LLM interface that provides If-Then plans based on short-form viewing patterns. Through this approach, we aim to explore the potential of a new HCI intervention model that can improve self-regulation abilities without coercive intervention and form healthy media usage habits.

2 Related Work

Research on the impact of short-form video platforms is being conducted following their rapid proliferation. One study presented evidence that short-form video viewing impairs executive control functions of the prefrontal cortex and negatively affects self-control [29]. Existing intervention methods have primarily relied on coercive restrictions but have limitations in that they can cause

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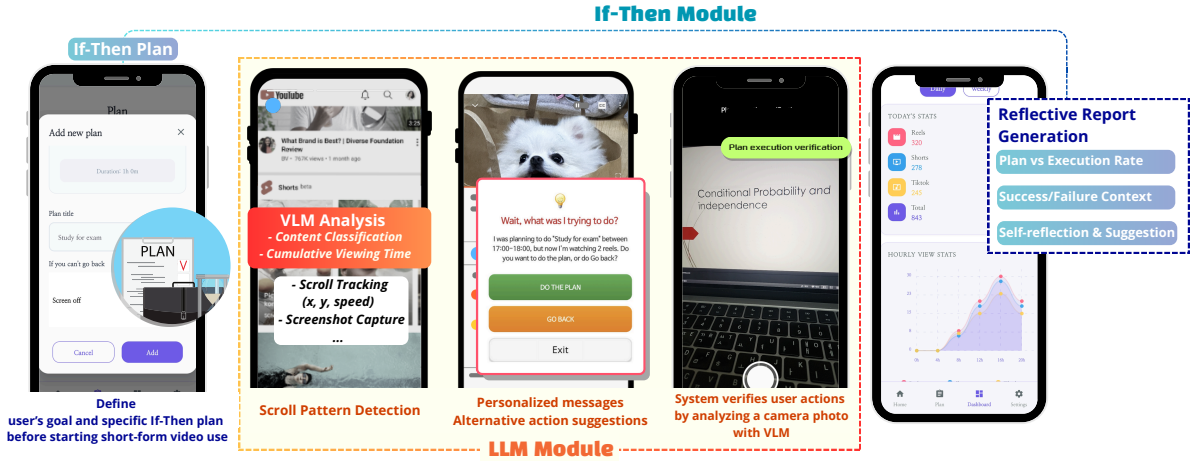


Figure 1: Overview of the SHIFT system workflow: When the app launches, the user sets personal goals and an If–Then plan. The dashboard visualizes viewing statistics, content distribution, and intervention–response rates. Upon reaching the 15-minute threshold, a personalized intervention overlay appears, allowing the user to carry out the planned behavior, switch to an alternative activity, or dismiss the overlay.

frustration and rather trigger reactance [1, 15]. Accordingly, the need for contextual interventions is emerging, but most studies have remained at the level of investigating contextual information [5, 15, 18] and failed to provide personalized specific behavioral change mechanisms [22, 28].

From a technical perspective, direct access to content is restricted due to enhanced privacy policies since Android 10 [11]. Under these constraints, Vision-Language Models (VLM) have shown the ability to accurately understand mobile UI and viewing types using only screenshots [6, 13, 27]. However, existing research has only focused on data collection and analysis without leading to actual behavioral change [23]. From a personalization technique perspective, recommendation systems show high performance in personalization but are designed to increase content consumption [14], which contradicts intervention purposes. Additionally, a theoretical foundation is needed that can convert intentions into actual behaviors along with personalized approaches [1].

In this context, If-Then theory presents a self-regulation strategy that converts goal intentions into actual behaviors through if-plans of “if situation Y occurs, then I will perform action Z” [10, 25]. Therefore, this study proposes an interface that inversely utilizes personalization techniques of recommendation systems to induce alternative behaviors for short-form viewing based on the If-Then theory.

3 Method

This study implemented SHIFT, an LLM-based adaptive interface for short-form video viewing intervention. SHIFT is a system that combines viewing goals with real-time behavioral patterns to generate personalized intervention messages. The overall workflow is presented in Figure 1.

VLM-based Content Analysis and Classification The Accessibility Service automatically activates in the background to monitor scroll coordinates and the foreground status in real time.

When short-form apps are inactive, only minimal app status checks are performed to minimize battery consumption. When content switching through scrolling is detected, screenshots are captured. The captured screens are analyzed through VLM to extract descriptions and key tags of the content being viewed. During this process, the content type, viewing time, and cumulative viewing count are recorded in logs. VLM [20, 21], which extracts a concise description and key tags of the content currently being viewed. When the user selects the planned or an alternative action, a photo taken by the camera is processed by the same VLM to verify that the chosen activity is actually being performed.

Prompt-based Intervention Message Generation Based on collected data, four prompting strategies [17] are applied: (1) GPT-4o [20, 21]–based VLM reconstructs extracted information; (2) generates goal achievement-focused messages based on plans and viewing patterns; (3) motivational recommendations based on pre-selected alternative activities and successful messages; and (4) embedding space integration. Through this, personalized text is generated.

Adaptive Intervention and Learning Based on existing research, when the 15-minute threshold is reached [24, 26], personalized intervention messages (current viewing statistics, recommended alternative activities) are presented as overlays through System Alert Window. All user responses are recorded in Firebase [12] to continuously learn individual intervention effectiveness.

4 Conclusion and Future Work

We recruited 30 participants who wanted to reduce their short-form viewing time. We conducted a 4-week user test using an ABAB (reapplication) design to verify its effectiveness. The first week was the baseline (A_1), the second week was the SHIFT intervention (B_1), the third week was the intervention withdrawal (A_2), and the fourth week was the reintervention (B_2). Viewing time was measured daily

for each week, yielding seven data points per phase. The experimental results clearly demonstrated the consistency and causality of the intervention effects through the ABAB design. Participants' average daily short-form viewing time changed from A_1 (2 hours 10 minutes) $\rightarrow B_1$ (1 hour 5 minutes) $\rightarrow A_2$ (1 hour 55 minutes) $\rightarrow B_2$ (1 hour 8 minutes), showing a consistent 50% reduction during the intervention ($p < 0.001$). The intervention success rate was 83% for B_1 and 81% for B_2 , demonstrating the reproducibility and reliability of the system ($p < 0.001$). This study demonstrated the potential of a digital well-being intervention system that combines viewing pattern-based personalization and If-then theory. In particular, the ABAB design enabled the causality of the intervention effect to be proven without a control group, and the consistent response pattern in the intervention-withdrawal-reintervention process ensured the reliability of the system. However, limitations exist, including the small sample size, inherent limitations of a single-group design, short-term observation, and homogeneity of the participant group.

Future studies will evaluate long-term effectiveness through longitudinal research spanning more than two months. Additionally, we plan to conduct comparative studies incorporating a randomized control group alongside an ABAB design extended to two weeks per phase. The goal is to validate effectiveness across diverse age groups and cultural backgrounds to ensure the system's universality. Future research will explore global applicability through expansion studies on overseas platforms such as Snapchat.

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